

claude-windows / fad100cc-2a38-4bb7-8ce8-40c750cb0086

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\fad100cc-2a38-4bb7-8ce8-40c750cb0086\subagents\agent-a9cc1abc62607d170.jsonl`  
- SHA-256: `9afc76fccdc5a71ae8cbbe83a59af108ce6e6863ee3a9be9594d00a95d17511f`  
- Source modified: `2026-06-22T09:20:37+00:00`  
- Imported at: `2026-07-05T16:48:22+00:00`  
- project: `subagents`  
- session_id: `fad100cc-2a38-4bb7-8ce8-40c750cb0086`
```

Transcript

1. user (2026-06-22T09:18:48.358Z)

Implement a SCOPED P0 (owner-confirmed "safe scope"): scrub personal machine paths from CODE docstrings + add a leak-scan CI guard. ****DO NOT**** rename the functional video names (eval_baselines keys, `rally\_seg\_eval.py::MULTICOURT\_CLIPS`, quality-log/comment names like "mahadevpura-1"/"kushagra") ? that rename is DEFERRED (it's coupled to the owner's gitignored manifest + data files + the training floor, and collides with an active session). Touch **ONLY** the files listed below.

WORK ONLY IN THIS WORKTREE (isolated; do NOT touch the main checkout or other worktrees):

C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-p0\
All paths relative to that root. Do NOT run git (no add/commit/push) ? create/edit files + run tests only; I handle git.

=== 1. SCRUB personal machine paths + the trivial non-functional name clutter in THESE code docstrings/examples ONLY ===

- `backend/eval/eval\_partial\_labels.py`: in the module docstring (~lines 23-24) replace the personal paths `C:/Users/avidu/AppData/Local/Temp/adarsh\_avi\_traj.csv` and `C:/Users/avidu/Projects/Annotation Setup/Collect/Adarsh and Avi.rallies.csv` with neutral placeholders (e.g. `output/<video>\_traj.csv` and `path/to/<labels>.rallies.csv`). The output-default at ~line 119 (`"adarsh\_avi\_vs\_human.csv"`) ? a neutral default like `"comparison\_vs\_human.csv"` (verify it is only a DEFAULT filename, not a functional key; if it's functional, leave it + report).
- `backend/eval/gemini\_refine.py`: in the module docstring (~lines 18-19) replace the example tokens `Adarsh.MP4`, `adarsh\_traj.csv`, `output/adarsh\_gemini.csv` with neutral placeholders (e.g. `<video>.MP4`, `<trajectory>.csv`, `output/<video>\_gemini.csv`).
- `README.md` (~line 281): replace `C:\Users\avidu\Videos\LargeTestVideo.MP4` with a neutral placeholder like `path/to/your/video.MP4`.

Do NOT edit any other file for names/paths (NOT eval_baselines/*.json, NOT QUALITY_ITERATIONS.md/NEXT_STEPS.md/CODE_MAP.md, NOT windowing.py/tracknet_runner.py/config models.py/fusion_hybrid.py comments, NOT rally_seg_eval.py, NOT scratch/ or docs/archives/). Those are the DEFERRED name-rename + hot files.

=== 2. ADD the leak-scan guard: tools/leak_scan.py ===

- A pure-logic, importable module + CLI (`python -m tools.leak\_scan`). It scans ****tracked text files**** under these roots only: `backend/`, `training/`, `tests/`, `scripts/`, `tools/`, `eval\_baselines/`. It EXCLUDES `tools/leak\_scan.py` itself AND its test file (they legitimately contain sample patterns), and skips binary/non-text files.
- Flags, with `file:line`: (a) a 32-hex-char token `b[0-9a-fA-F]{32}b` (the `video\_id`/MD5 shape ? note golden fixtures use 16-hex `fx\_` slugs and `gt\_hash` is 12-hex, which must NOT trip it); (b) a personal absolute user path ? `[Cc]:[\\/]Users[\\/]<name>` or `/(?:(home|Users)/<name>/` ? where `<name>` is a real username, i.e. NOT one of an allowlist of placeholders (`<name>`, `<user>`, `<username>`, `you`, `youruser`, `yourname`); (c) OPTIONAL: if a gitignored private name-list file exists (path via `--names <file>`, default `.leakscan\_private`), also flag any of those tokens (case-insensitive, word-ish boundary).
- Exit 1 if any finding (printing each `file:line: <reason>`), exit 0 if clean. Support `--root <dir>` (default repo root, resolved from `\_\_file\_\_`) and `--names <file>`. Keep it dependency-

free (stdlib only; discover tracked files via `git ls-files` through subprocess, OR walk the roots and skip common ignores ? prefer `git ls-files` so it only scans tracked files).

- Add `.leakscan\_private` to `.gitignore` (the private name-list is never committed).
- `tests/test\_leak\_scan.py`: unit tests over a tmp tree ? a personal `C:\Users\realuser\...\` path is flagged, a placeholder `C:\Users\<name>\...\` is NOT, a 32-hex MD5 is flagged, a 16-hex `fx\_` slug and 12-hex hash are NOT, the private-name-list flags a listed token, a clean tree returns exit 0, and the scanner excludes itself. Use tmp dirs (no real PII committed in the test ? build sample strings at runtime, e.g. concatenate to avoid the test file itself tripping a scan).

=== 3. CI: .github/workflows/ci.yml ===

- Add a `leak-scan` job (ubuntu, fast) that runs `python -m tools.leak\_scan` (exit non-zero fails the job). It is a structural guard (MD5 + personal paths); it does NOT need the private name-list (CI won't have it).

=== 4. DOC: docs/GOLDEN_REGRESSION_FIXTURES.md ===

- \$7 P0 row ? status `?` (partial): "paths scrubbed from code docstrings + leak-scan CI guard added (this PR); the FUNCTIONAL video-name rename (eval_baselines keys / MULTICOURT_CLIPS / quality-log names) is DEFERRED per the owner's safe-scope decision 2026-06-16 ? it's coupled to the gitignored manifest + data files + the training floor and collides with the active quality track; revisit before open-sourcing." Add a \$7 Progress-log row for this PR. Keep it honest.

=== VERIFY (run in the worktree; report verbatim) ===

1. `python -m tools.leak\_scan; echo EXIT=\$?` ? MUST exit 0 (clean) after your scrub. If it flags something you were told NOT to scrub (e.g. a name in eval_baselines), that means the scan scope is too broad ? NARROW the scan to not flag the DEFERRED names (it should only flag MD5 + personal paths, not the video names), then re-run. Report what it found, if anything.
2. `python -m pytest -q --cov=backend --cov=training --cov-fail-under=70` ? FULL suite green; report pass/skip + coverage (must stay >= 80.7%).
3. `ruff check .` and `mypy backend training` ? clean (the new tool must be typed + lint-clean). Report: files changed, verbatim verify output, and any item you could not do + why. Do NOT run git.

2. assistant (2026-06-22T09:18:50.550Z)

I'll start by reading the target files to understand their current state, then make the scoped changes.

3. user (2026-06-22T09:18:51.993Z)

```
1      """Evaluate the rally pipeline against PARTIAL human golden labels.
2
3      The VLC rally-annotator produces a human-rated CSV (rally_number,start_time,end_time,
4      ending_reason,sport). While a video is only partially tagged, the labels cover a
5      PREFIX of the video ? every rally up to some point is marked, and everything after is
6      simply un-watched (not "no rally there"). Scoring naively against such labels is wrong:
7      the detector's *correct* rallies in the un-labeled tail would be counted as false
8      positives and crush precision.
9
10     This harness fixes that by restricting the evaluation to the LABELED WINDOW
11     [window_start, window_end] (window_end defaults to the last labeled rally's end) and
12     dropping predicted segments that fall outside it. Within that window the human labels
13     ARE complete, so precision/recall/F1 are honest.
14
15     It reuses everything from the existing eval stack (no new windowing/metrics/gate logic):
16     - calibrate_wasb.make_predict_fn / load_golden_gts / BASELINE
17     - pipeline.detectors.rally_gate.apply_gate   (net-crossing "gate A")
18     - eval.metrics.match_intervals / score
19     - export_wasb_segments.build_comparison / write_comparison_csv   (per-rally detail CSV)
20
21     Run:
22     python -m backend.eval.eval_partial_labels \
23         --trajectory C:/Users/avidu/AppData/Local/Temp/adarsh_avi_traj.csv \
24         --labels "C:/Users/avidu/Projects/Annotation Setup/Collect/Adarsh and
25     Avi.rallies.csv" \
```

```

25         --fps 59.94 --frame-width 1920
26     """
27     from __future__ import annotations
28
29     import argparse
30     import os.path as osp
31     from typing import List, Optional, Tuple
32
33     from backend.eval.calibrate_wasb import make_predict_fn, load_golden_gts, BASELINE
34     from backend.eval.metrics import score
35     from backend.eval.export_wasb_segments import build_comparison, write_comparison_csv
36     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
37     from backend.pipeline.detectors import rally_gate
38
39     Interval = Tuple[float, float]
40     Config = Tuple[float, float, float]
41
42     # Operating points to compare. The tuned config wins on its fit-video but over-segments
43     # and does not transfer; baseline (+ net-crossing gate) is the stronger general point.
44     TUNED: Config = (0.05, 1.0, 1.0)
45
46
47     def restrict_to_window(intervals: List[Interval], w0: float, w1: float) ->
List[Interval]:
48         """Keep intervals that overlap the labeled window [w0, w1].
49
50         A predicted segment is evaluated iff it overlaps the labeled region; segments wholly
51         in the un-labeled tail (start >= w1) or before the window (end <= w0) are dropped so
52         they are neither rewarded nor penalized. GTs are assumed already inside the window.
53         """
54         return [(s, e) for (s, e) in intervals if e > w0 and s < w1]
55
56
57     def _eval_point(predict_fn, cfg: Config, points, fps: float, gts: List[Interval],
58                     window: Tuple[float, float], gate_crossings: int, iou: float) -> dict:
59         """Score one (config, gate) operating point inside the labeled window."""
60         preds = predict_fn(cfg)
61         if gate_crossings > 0 and points:
62             preds = rally_gate.apply_gate(points, preds, fps, min_crossings=gate_crossings)
63             preds = restrict_to_window(preds, *window)
64             sc = score(preds, gts, iou)
65             return {"cfg": cfg, "gate": gate_crossings, "n_pred": len(preds), "score": sc,
66 "preds": preds}
67
68     def run(trajecory_csv: str, labels_csv: str, fps: float, frame_width: float,
69            window_start: float = 0.0, window_end: Optional[float] = None,
70            gate_crossings: int = 2, iou: float = 0.5,
71            comparison_out: Optional[str] = None) -> dict:
72         gts_all = load_golden_gts(labels_csv)
73         if not gts_all:
74             raise SystemExit(f"no usable human rallies in {labels_csv}")
75         if window_end is None:
76             window_end = max(e for _, e in gts_all)
77         window = (window_start, window_end)
78         gts = restrict_to_window(gts_all, *window)
79
80         predict_fn = make_predict_fn(trajecory_csv, fps, frame_width)
81         points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
82         traj_end = (max(p.frame for p in points) / fps) if points else 0.0
83
84         print("=== Pipeline vs PARTIAL human labels ===")
85         print(f"labels: {labels_csv}")
86         print(f" {len(gts_all)} human rallies; labeled window [{window_start:.1f}s,
{window_end:.1f}s] ")

```

```

87         f"-> {len(gts)} rallies in-window")
88     print(f"trajectory covers ~0-{{traj_end:.1f}}s ")
89     f"({{'covers the window' if traj_end >= window_end - 1 else 'WARNING: shorter
than window'}})")
90     print(f"IoU>={{iou}}; net-crossing gate K={{gate_crossings}}\n")
91
92     # Compare the standard operating points: baseline / baseline+gate / tuned /
tuned+gate.
93     points_for_gate = points
94     candidates = [
95         ("baseline", BASELINE, 0),
96         (f"baseline+gate{{gate_crossings}}", BASELINE, gate_crossings),
97         ("tuned", TUNED, 0),
98         (f"tuned+gate{{gate_crossings}}", TUNED, gate_crossings),
99     ]
100     results = []
101     header = f"{'operating point':<18} {'pred':>4} {'TP':>3} {'FN':>3} {'FP':>3} " \
102         f"{'prec':>6} {'recall':>6} {'F1':>6} {'mIoU':>6} {'sErr':>6} {'eErr':>6}"
103     print(header)
104     print("-" * len(header))
105     for name, cfg, gate in candidates:
106         r = _eval_point(predict_fn, cfg, points_for_gate, fps, gts, window, gate, iou)
107         sc = r["score"]
108         r["name"] = name
109         results.append(r)
110         print(f"{{name:<18}} {{r['n_pred']:>4}} {{sc.true_positives:>3}}
{{sc.false_negatives:>3}} "
111             f"{{sc.false_positives:>3}} {{sc.precision:>6.3f}} {{sc.recall:>6.3f}}
{{sc.f1:>6.3f}} "
112             f"{{sc.mean_iou:>6.3f}} {{sc.mean_start_error:>6.2f}}
{{sc.mean_end_error:>6.2f}}")
113
114     best = max(results, key=lambda r: r["score"].f1)
115     print(f"\nbest operating point by F1: {{best['name']}} (F1 {{best['score'].f1:.3f}})")
116
117     # Per-rally detail for the best point: which human rallies matched/missed + extra
fires.
118     comp = build_comparison(best["preds"], gts, iou)
119     comp_out = comparison_out or osp.join("output", "adarsh_avi_vs_human.csv")
120     write_comparison_csv(comp_out, comp)
121     print(f"per-rally detail ({{best['name']}}) -> {{comp_out}}")
122
123     return {
124         "window": window, "n_human_in_window": len(gts),
125         "results": [{"name": r["name"], "cfg": r["cfg"], "gate": r["gate"],
126                     "n_pred": r["n_pred"], **r["score"].as_dict()} for r in results],
127         "best": best["name"], "comparison_csv": comp_out,
128     }
129
130

```

4. user (2026-06-22T09:18:52.929Z)

```

1     """Refine WASB rally candidates with Gemini (verify boundaries + reject false fires).
2
3     WASB (shuttle-only) over-fires on between-point service feeds. This drives the
4     EXISTING Gemini provider over the SHORT WASB candidate clips (not the whole video):
5     each candidate window is cut to a small padded clip and handed to Gemini, which ?
6     seeing the actual footage ? returns the true rally inside it (tight boundaries) or
7     nothing (it was a toss / dead time). Cheap because uploads are seconds-long clips,
8     not GB-scale chunks; high-precision because Gemini judges each candidate visually.
9
10    Reuses (no reinvented wheels): providers.GeminiProvider (via provider_registry),
11    sports.badminton prompt (sport_registry), utils.video.extract_video_chunk,
12    calibrate_wasb.make_predict_fn, export_wasb_segments.write_segments_csv.

```

```

13
14     This mirrors wasb_hybrid's Phase-3 handoff but runs standalone on a cached
15     trajectory (no WASB re-run, no DB) ? an eval/tuning driver, not the live pipeline.
16
17     Run (GEMINI_API_KEY in env):
18         python -m backend.eval.gemini_refine --video Adarsh.MP4 --trajectory adarsh_traj.csv
19         \
20             --fps 59.94 --frame-width 1920 --out output/adarsh_gemini.csv [--limit 4]
21     """
22     from __future__ import annotations
23
24     import os
25     import os.path as osp
26     import time
27     import argparse
28     import concurrent.futures
29     from typing import List, Tuple, Optional
30
31     from backend.eval.calibrate_wasb import make_predict_fn, BASELINE
32     from backend.eval.export_wasb_segments import TUNED, write_segments_csv, seconds_to_mmss
33     from backend.utils.video import get_video_duration, extract_video_chunk
34     from backend.sports import sport_registry
35     from backend.providers import provider_registry
36
37     Interval = Tuple[float, float]
38
39     def merge_overlaps(intervals: List[Interval], gap_tol: float = 0.0) -> List[Interval]:
40         """Union of overlapping intervals, optionally stitching across a small gap.

```

5. user (2026-06-22T09:18:53.559Z)

```

270     ```bash
271     python -u -m backend.main --video-path "C:/path/to/video.mp4"
272     ```
273     * **Video Seeking Overhead**: Seeking to random frames in high-bitrate long H.265/HEVC
274     videos using OpenCV's `cap.set(cv2.CAP_PROP_POS_FRAMES)` incurs massive I/O decoding delays. The
275     `motion` segmenter reads sequentially and skips frame decodes using container-level
276     `cap.grab()`, which delivers a **100x speedup** on multi-gigabyte raw files. `yolo_hybrid`
277     sidesteps the problem entirely by extracting fixed chunks via ffmpeg first.
278
279     ### Q5: Why did the video fail with "No keyframes / packet timestamps could be
280     extracted"?
281     Some H.265/HEVC streams (e.g. GoPro footage) encode frames as GDR or CRA keyframes.
282     Standard Ffmpeg frame-level skipping (`-skip_frame nokey`) fails to identify them, returning
283     empty logs. We resolved this by extracting packet indexes (`packet=pts_time,flags`) at the
284     container demuxer level instead, which is incredibly robust and executes instantly without
285     decoding a single pixel.
286
287     ### Q6: How can I quickly test the indexing without waiting to parse an entire multi-GB
288     video file?
289     You can pass the `--max-frames <N>` argument in CLI or `max_frames` in the POST API
290     payload. This restricts the segmenter frame loop to process only the first $N$ sub-sampled
291     frames, enabling you to test motion filters and court relevance checks in seconds:
292     ```bash
293     python -m backend.main --video-path "C:\Users\avidu\Videos\LargeTestVideo.MP4" --max-
294     frames 300
295     ```
296     *(At 5 sub-sampled frames per second, `--max-frames 300` analyzes exactly the first 60
297     seconds of video).*
298
299     ### Q7: How do I configure and run an AI-powered segmenter (Gemini / YOLO-hybrid)?
300     Both `yolo_hybrid` (default) and `gemini` need an API key for the AI provider. To use
301     Google Gemini:
302     1. Create a local `.env` file in the project root (this is already in `.gitignore` to

```

```

prevent secret leaks).
288     2. Set your Google AI Studio API key in the file:
289         ```env
290         GEMINI_API_KEY=your_key_here
291         ```
292     3. Run with whichever segmenter you want:
293         ```bash
294         # Default: cheap candidate filtering with YOLO, then precise boundaries via Gemini

```

6. assistant (2026-06-22T09:18:56.287Z)

Now let me look at the supporting files: the existing CI workflow, the golden fixtures doc, gitignore, and golden fixture patterns to understand the hash shapes.

7. user (2026-06-22T09:18:57.256Z)

```

1     name: CI
2
3     on:
4         push:
5             branches: [master]
6         pull_request:
7
8     permissions:
9         contents: read
10
11    concurrency:
12        group: ci-${{ github.ref }}
13        cancel-in-progress: true
14
15    jobs:
16        # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
17        # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
18        # tracked .py, including scripts/ and training/, which the backend-only
19        # py_compile smoke sweep does not reach.
20        lint:
21            name: Ruff lint
22            runs-on: ubuntu-latest
23            steps:
24                - uses: actions/checkout@v6
25                - uses: actions/setup-python@v6
26                  with:
27                      python-version: "3.13"
28                - name: Install ruff
29                  run: python -m pip install ruff
30                - name: Ruff check
31                  run: ruff check . --output-format github
32
33        # Type gate (mypy.ini at repo root): lenient green baseline with an explicit
34        # per-module quarantine list ? ratchet by removing entries as modules clean
35        # up. Typed core deps are installed so mypy sees their real signatures
36        # (without them ignore_missing_imports would Any-stub half the checks).
37        typecheck:
38            name: Mypy (lenient baseline)
39            runs-on: ubuntu-latest
40            steps:
41                - uses: actions/checkout@v6
42                - uses: actions/setup-python@v6
43                  with:
44                      python-version: "3.13"
45                - name: Install mypy + typed core deps
46                  run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-
genai numpy
47                - name: Mypy
48                  run: mypy backend training

```

```

49
50 # Fast guard against the 7fbb0 class of breakage: a SyntaxError or
51 # module-level import error shipping while the suite stays green.
52 # Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
53 # on machines without the GPU stack (test_import_smoke stubs what's absent).
54 import-smoke:
55     name: Syntax sweep + import smoke (no GPU stack)
56     runs-on: ubuntu-latest
57     steps:
58     - uses: actions/checkout@v6
59     - uses: actions/setup-python@v6
60       with:
61         python-version: "3.13"
62     - name: Install minimal deps
63       run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai
64     - name: Run import smoke tests
65       run: python -m pytest tests/test_import_smoke.py -v
66
67 full-suite:
68     name: Full test suite (offline, mocked)
69     runs-on: ubuntu-latest
70     steps:
71     - uses: actions/checkout@v6
72     - uses: actions/setup-python@v6
73       with:
74         python-version: "3.13"
75         cache: pip
76     - name: System libs for opencv-python
77       run: |
78         sudo apt-get update
79         sudo apt-get install -y libgl1 libglu1-mesa-dev || sudo apt-get install -y
libgl1 libglu1-mesa-dev
80     - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
81       # torch/torchvision are NOT in pyproject deps ? their CPU/CUDA wheels
82       # need an out-of-band index that PEP 621 can't express. Keep installing
83       # them separately here, before the editable install pulls the rest.
84       run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
85     - name: Install package (editable) + dev tooling
86       # Editable PEP 621 install: runtime deps + the `dev` group
87       # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
88       run: python -m pip install -e .[dev]
89       # obsreport (private, soft dep) is absent here: reporting tests skip via
90       # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
91       # is therefore calibrated against THIS lane's number (local runs with
92       # obsreport installed measure a little higher).
93       # Floor calibrated 2026-06-11 against this lane's measured 72% (local
94       # with obsreport: 73%). Ratchet it UP as coverage grows; never down
95       # without an owner decision.
96     - name: Run full suite (with coverage floor)
97       run: python run_all_tests.py --cov=backend --cov=training --cov-report=term
--cov-fail-under=70
98
99 # Cross-OS determinism guard for the committed golden regression fixtures
100 # (docs/GOLDEN_REGRESSION_FIXTURES.md). The CONTROL replay path is pure-stdlib
101 # (only numpy, no cv2/torch), so this is a FAST, focused job ? it does NOT run
102 # the full suite x3 OSes. It replays the Windows/Linux/macOS runner against the
103 # committed (parse-compared) snapshots, catching any cross-OS float drift beyond
104 # the 1e-6 tolerance before it can flake a contributor's PR on another machine.
105 golden-crossos:
106     name: Golden fixtures cross-OS (${ matrix.os })
107     strategy:
108       fail-fast: false
109     matrix:

```

```

110     os: [ubuntu-latest, windows-latest, macos-latest]
111 runs-on: ${ matrix.os }
112 steps:
113   - uses: actions/checkout@v6
114   - uses: actions/setup-python@v6
115     with:
116       python-version: "3.13"
117   - name: Install minimal deps (no GPU stack ? the CONTROL replay needs only numpy)
118     run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai numpy
119   - name: Golden-fixtures characterization (cross-OS determinism guard)
120     # Also runs the freeze-real / time-origin-reset / M2 refusal-gate tests
(synthetic /
121     # tmp_path only ? no GPU/network) so the reset metric-invariance (fix #2) and
the
122     # de-attribution paths get cross-OS coverage and surface any win/mac float
drift.
123     run: |
124       python -m backend.eval.golden_fixtures --check
125       python -m pytest tests/test_golden_fixtures.py
tests/test_golden_real_fixtures.py -q
126
127     # Commercial-clean licensing guardrail (MIT/Apache/BSD only ? PACKAGING_PLAN.md P0d).
Scans the
128     # RUNTIME dependency closure (`pip install .` ? no torch/dev) and fails on any
copyleft
129     # (GPL/AGPL/LGPL/SSPL). torch (BSD-3) is installed out-of-band and is not in this
closure. This
130     # automates a guardrail that was human-only; the report is printed for review either
way.
131   license-scan:
132     name: Dependency license scan (no copyleft in runtime deps)
133     runs-on: ubuntu-latest
134     steps:
135       - uses: actions/checkout@v6
136       - uses: actions/setup-python@v6
137       with:
138         python-version: "3.13"
139       - name: Install runtime deps + pip-licenses
140       run: |
141         python -m pip install .
142         python -m pip install pip-licenses
143       - name: Report dependency licenses
144       run: python -m piplicenses --format=markdown --with-urls --order=license
145       - name: Fail on any copyleft (GPL/AGPL/LGPL/SSPL) in the runtime closure
146       run: |
147         if python -m piplicenses --format=plain | grep -iE "GPL|SSPL"; then
148           echo "::error::A runtime dependency carries a copyleft (GPL/AGPL/LGPL/SSPL)
license ? the guardrail is MIT/Apache/BSD only. Review the report above."
149           exit 1
150         fi
151       echo "OK: no GPL/AGPL/LGPL/SSPL in the runtime dependency closure."
152
153     # Built-wheel smoke (PACKAGING_PLAN.md P0d): build the wheel, install it NON-editable
from a
154     # non-repo cwd, and launch the console script. Catches packaging regressions the
source suite
155     # can't ? namespace-package resolution (backend has no __init__.py), missing data
files, hidden
156     # imports. (Validated locally: a clean-venv wheel install + CPU torch runs `indexer
--help`.)
157   build-artifact:
158     name: Built wheel smoke (install + launch)
159     runs-on: ubuntu-latest
160     steps:

```



```

161         - uses: actions/checkout@v6
162         - uses: actions/setup-python@v6
163         with:
164             python-version: "3.13"
165         - name: System libs for opencv-python
166           run: |
167             sudo apt-get update
168             sudo apt-get install -y libgl1 libglib2.0-0t64 || sudo apt-get install -y
libgl1 libglib2.0-0
169         - name: Build the wheel
170           run: |
171             python -m pip install build
172             python -m build --wheel
173         - name: Install the built wheel + CPU torch (out-of-band index)
174           run: |
175             python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
176             python -m pip install dist/*.whl
177         - name: Launch the installed console script from a non-repo dir
178           run: |
179             cd /tmp
180             indexer --help
181             python -c "import backend, training, backend.utils.app_home; print('wheel
import OK')"
182

```

8. user (2026-06-22T09:18:57.821Z)

```

1      # Byte-compiled / optimized / DLL files
2      __pycache__/
3      *.py[cod]
4      *$py.class
5      *.pt
6
7      # Local environment secrets
8      .env
9      *.env
10     .env.*
11
12     # Provenance stamp baked into a deploy tarball at build time (deploy/gcp/README.md §4);
13     # never a tracked source file ? generated per-deploy, must not ride into a commit.
14     /GIT_SHA
15
16     # C extensions
17     *.so
18
19     # Distribution / packaging
20     .Python
21     build/
22     develop-eggs/
23     dist/
24     downloads/
25     eggs/
26     .eggs/
27     lib/
28     lib64/
29     parts/
30     sdist/
31     var/
32     wheels/
33     share/python-wheels/
34     *.egg-info/
35     .installed.cfg
36     *.egg
37     MANIFEST

```

```

38
39 # SQLite databases & Local output files
40 *.db
41 *.db-journal
42 *.db-wal
43 output/
44 output_dir/
45 backend/static/highlights_*.mp4
46 backend/static/*_compiled.mp4
47
48 # OS generated files
49 .DS_Store
50 Thumbs.db
51 desktop.ini
52
53 # Coverage data
54 .coverage
55 .coverage.*
56 htmlcov/
57 .pytest_cache/
58
59 # Local Claude Code config (machine-specific)
60 .claude/
61 .clocov-e-r-a-g-e
62
63
64 # Throwaway repro scripts + PR-body / handoff scratch notes (already excluded from
65 # ruff + pytest; never meant to be committed ? keep them out of `git status` too).
66 scratch/
67 .venv/
68

```

9. assistant (2026-06-22T09:19:01.277Z)

Now let me look at the golden fixtures doc §7 and the golden fixture patterns (16-hex fx_slugs, 12-hex gt_hash) to ensure my regexes don't trip on them.

10. user (2026-06-22T09:19:02.652Z)

```

1 # Golden Regression Fixtures ? video-free, non-attributable regression suite
2
3 > **Status:** `IN PROGRESS` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 .
**Last updated:** 2026-06-16 . **Legal:** owner **accepted the researched risk** 2026-06-16
(condition §3 stand; this is the owner's informed acceptance, **not** a retained-counsel
opinion)
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE, per archive policy)
5 > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
6 > **Relation to existing docs:** EXTENDS (does not supersede)
`docs/GOLDEN_DATA_SHARING.md` (#175); reuses the eval stack documented in `docs/EVALUATION.md` /
`docs/EXPERIMENT_HARNESS.md`; quality floor ties to `docs/QUALITY_ITERATIONS.md`.
7 > **Honesty note:** §3 (Legal) is **researched general information, NOT legal advice** ?
it gates on retained counsel (§8). §2/§4/§7 mark which claims are **[verified]** against code vs
**[design]** proposals.
8
9 ---
10
11 ## 0. TL;DR
12
13 Let a contributor on **another machine with no video, GPU, WSL, or DB** run the rally-
segmentation regression suite from **committed structured files**, so refactors and non-quality
PRs get a red/green side-effect signal. Two layers, built once and grown by accretion:
14
15 1. **Mechanics** ? freeze the deterministic pipeline at the **post-detector trajectory

```

```

seam** and replay it through the *already-shipped* eval stack (`experiment.py` / `metrics.py` /
`windowing.py` / `regression.py`), with two auto-discovered gates: a
**characterization/snapshot** gate (catches any behaviour change) and a **quality-floor** gate
(catches quality drops).
16      2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, owner-source,
biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-
repo. Non-attributability comes from **deletion-at-source**, not encryption. ? "Minimized" means
**stream minimization** (shuttle-only, no person/audio/pose) ? **NOT** reduction of the shuttle
stream to ~5 scalars: the de-attributed per-frame `trajectory.csv` IS committed (see §4c + its
risk-acceptance re-confirmation note).
17
18      **The convergence that makes this clean:** the *privacy* answer (publish shuttle-only,
drop person/pose streams) and the *determinism* answer (freeze only the `CONTROL` path ? which
runs **no numpy execution**, keeping the numpy/BLAS learned scorer off the cross-machine gate)
point at the **same minimal public tier**. That coincidence is the spine of the design. (numpy
is still *imported* transitively via `experiment` and must be installed; the CONTROL replay just
never calls into it.)
19
20      **Two findings already actionable, independent of the rest:**
21      - ? **Live de-attribution bug** *(verified)*: real player/venue names are committed
today in `eval_baselines/heuristic_lovo_n6.json`,
`eval_baselines/gemini_wrapper_2026-06-10.json`, and `backend/eval/eval_partial_labels.py` (the
last also has a personal `C:/Users/avidu/...` path) ? and in git history. ? **Deliverable P0.**
22      - ? **Determinism trap** *(verified, corroborated by two independent red-teams)*: the
learned variants (`shuttle_only`, `fusion`) train a numpy/OpenBLAS `DYNAMIC_ARCH` logistic
regression whose decisions flip near the 0.5 threshold across CPUs ? would false-RED on a
contributor's machine. ? the cross-machine gate must be **CONTROL/static-only**.
23
24      ---
25
26      ## 1. Problem & goal
27
28      The rally-quality measurement is split across two machines by capability
(`docs/GOLDEN_DATA_SHARING.md`):
29      - `backend/eval/fusion_golden.py` **extracts** `_golden_features.json` ? needs the
**video + trajectory CSV + labels + GPU/WSL** (GPU box only).
30      - `backend/eval/fusion_compare.py` / `backend/eval/ablation.py` **re-score** those JSONs
on **pure CPU, no video** ? any machine.
31
32      The CPU re-scoring is *already* video-free, but the inputs live in gitignored `output/`
/ private OneDrive, so the regression tests that depend on them **skip everywhere**
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`). The
committed `eval_baselines/` baselines exist but `regression.py` still scores DB-stored
predictions (needs a pipeline run on the video).
33
34      **Goal:** commit a tiny, version-tagged, **numeric-only, non-attributable** per-fixture
corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression
signal with **zero video/GPU/DB**, while never leaking attributable or sensitive data.
35
36      ---
37
38      ## 2. Decisions locked
39
40      | # | Decision | Source | Implication |
41      |---|-----|-----|-----|
42      | D1 | **Corpus is owner-recorded / owned** | owner answer 2026-06-16 | Legal analysis
takes **Fork A** (clean): no third-party copyright/ToS infringement; concern is trade-
secret/moat + privacy. Strongest non-attributability (no public reference to correlate against).
|
43      | D2 | **Repo private now, may open-source later** | owner answer 2026-06-16 | Design
for **public-grade hygiene from day one** ? git history is permanent; nothing scrubable-later. |
44      | D3 | **Adults, consent not formalized** | owner answer 2026-06-16 | **Person/pose/gait
streams stay OUT of the public tier** (DPDP/GDPR derived-personal-data risk without consent).
Public tier = **shuttle + abstract intervals + non-biometric features only**. |
45      | D4 | **Freeze seam = post-detector trajectory** *(design; SHIPPED differs ? see ?)* |

```

workflow-1 synthesis | Widest deterministic surface; reuse the shipped eval stack. ?

****Correction (audit 2026-06-16):**** the shipped freeze ****commits** the de-attributed per-frame `(Frame,Visibility,X,Y)` `trajectory.csv` ****** (shuttle-only, time-origin reset) ? it is ****NOT**** minimized to ~5 aggregate scalars. See §4c (and its risk-acceptance re-confirmation note). |

46 | D5 | ****Cross-machine gate is CONTROL/static-only**** *(design)* | both red-teams [verified cause] | Learned variants (numpy/BLAS) are ****owner-box/Tier-1 only****, asserted with tolerance, never as a cross-machine byte-snapshot. |

47 | D6 | ****Non-attributability = deletion-at-source, encryption = public-only barrier**** *(design)* | workflow-2 synthesis | A contributor who runs the tests necessarily reads plaintext+key (DRM/analog-hole). So we ****delete**** attribution, not hide it. Encryption only stops the no-key public. |

48 | D7 | ****Two tiers**** *(design)* | workflow-2 synthesis | ****Tier-0**** public/no-key (shuttle-only, de-attributed). ****Tier-1**** key-gated/out-of-repo (person/audio, full corpus), ****skips cleanly when absent****. |

49

50 ---

51

52 ## 3. Legal foundation (researched ? gates on counsel, §8)

53

54 ****Core verdict (settled across IN/US/EU):**** the derived numbers ? per-frame shuttle (x,y), rally start/end intervals, audio/court/aggregate-person feature values ? ****do NOT inherit** the source video's copyright. ****** They are uncopyrightable *facts/measurements*, not authored expression, and not a §101 "derivative work."

55

56 | Jurisdiction | Copyright of the data | Database right | Contract / ToS | Net |

57 |---|---|---|---|---|

58 | ****India**** (primary) | Not copyrightable as facts (*EBC v. D.B. Modak*); but compilations get thin protection on a low "skill/labour" bar (*Burlington*, *OLX v. Padawan*) ? only bites if ****lifted from a 3rd-party dataset****, not self-derived | ****None**** (no sui-generis right) ? a real permissiveness edge | Anti-scraping ToS (untested in court) + ****IT Act s.43/s.66**** civil/criminal exposure for 3rd-party ingestion | ****Owned-source + minors-excluded + no person/pose = clean**** |

59 | ****US**** | Not copyrightable (*Feist*; §102(b); *NBA v. Motorola*) | None (*Feist* killed sweat-of-the-brow) | ****Dominant risk.**** ToS enforceable where copyright/CFAA fail (*ProCD*, *hiQ* ? injunction + ****forced deletion****). Remedy for redistribution is takedown, not nominal damages | Publishable as facts ****iff**** owned/permissioned source + no barring ToS + no person/biometric/minor stream |

60 | ****EU / UK**** | Not copyrightable (*Football Dataco v. Yahoo*) | ****Genuinely contestable.**** *Football Dataco v. Sportradar*: recording pre-existing on-court facts = ****obtaining****, not "creating" ? our "we created it" premise is ****overconfident****; harm test = *CV-Online Latvia* | Ryanair v. PR Aviation: ToS can ban extraction/redistribution even with no IP right | Get EU+UK advice before redistributing numeric streams at scale |

61

62 ****Provenance is the decisive fork.**** ****Fork A (owner-recorded, unpublished)**** ? the only path safe to commit publicly; collapses to a trade-secret *choice* (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 ? ****not**** US DTSA) + privacy. ****Fork B (broadcast/YouTube/scraped)**** ? do ****not**** commit; the extraction act needs a license or now-****contested**** US fair use (*Thomson Reuters v. Ross* 2025; USCO May-2025), and unlawful acquisition independently sinks it (*Bartz v. Anthropic*, ~\$1.5B settlement). Caveat: owner footage ****shot at a third party's event**** (tournament/club) carries ticket/venue/organiser/broadcaster terms ? treat as Fork B until cleared.

63

64 ****Publish-by-stream (privacy):****

65 - ? ****Publish-OK:**** shuttle (x,y)+Visibility (inanimate object), abstract rally/GT ****intervals**** (identity-stripped), non-biometric audio/court/aggregate-person features detached from identity.

66 - ?? ****Conditional (effectively never in v1):**** person bounding boxes that *persistently single out a player* ? flips to GDPR Art.9 the moment it identifies. This architecture is prone to that flip ? treat per-player tracking as a tripwire.

67 - ? ****Never publish:**** pose/skeleton/****gait**** (re-identifies at 80?95% even with faces blurred ? *Weiss et al. 2025*), face crops, any named-player?movement mapping. Matches the existing "gait/biometrics strictly future (GDPR Art.9)" guardrail.

68 - ? ****Minors flip everything:**** DPDP "child" = under 18; s.9(3) ****prohibits behavioural monitoring**** (the Rule-12 exemptions don't cover this project). A persistent per-player signature on a minor is, conservatively, prohibited. ****Exclude minors**** from any persisted

stream beyond inanimate shuttle/interval data.

69

70 ****Non-attributability is relative, not absolute.**** A per-video trajectory is a fingerprint (de Montjoye: 4 points ? 95% re-ID). It's only non-attributable because the ****Fork-A source stays unpublished**** (no public reference to correlate against). Per ***EDPS v. SRB*** (CJEU C-413/23 P, 2025) it ****remains personal data to the holder**** who can re-run the detector. So: conditional, revocable (publish the source and the link snaps back), forward-looking (re-assess as the public-video environment grows). Never market it as "anonymous."

71

72 ***Key authorities:** Feist 499 U.S. 340; NBA v. Motorola 105 F.3d 841; 17 U.S.C. §101/§102(b); Sega v. Accolade 977 F.2d 1510; Football Dataco v. Yahoo (C-604/10) & v. Sportradar (C-173/11); BHB v. William Hill (C-203/02); CV-Online Latvia (C-762/19); Ryanair v. PR Aviation (C-30/14); Thomson Reuters v. Ross (D. Del. 2025); EBC v. D.B. Modak; DPDP Act 2023 s.3/s.9 + Rules 2025; EDPS v. SRB (C-413/23 P). Full URLs in the research artifact (§10).

73

74 ---

75

76 **## 4. Architecture (design)**

77

78 **### 4a. Freeze seam + reuse map**

79 Freeze at the ****trajectory CSV ? CONTROL-path replay****. One shared helper ``backend/eval/golden_fixtures.py::replay_fixture()`` is the ***single*** code path used by both the freeze tool and both gates. As ****shipped****, that path is (per the module docstring):

80 ``parse_trajectory_csv ? distill_local.build_candidates(proposal=(vt,mg,mwd)) ? VideoCandidates(5 shuttle cols only) ? experiment.predict(CONTROL) [keep all ? merge_overlaps] ? metrics.score``.

81 Reuses ****verbatim****: ``metrics.score/interval_iou`, `experiment.predict/CONTROL/VideoCandidates`, `distill_local.build_candidates`, `tracknet_runner.parse_trajectory_csv`/`trajectory_to_action_windows`, `rally_gate` net-crossings (inside `build_candidates`), `gt_loader.load_gt_intervals` (strict), `regression.gt_hash`, `eval_baselines/` committed-baseline precedent.`

82 > ? ****Correction (audit 2026-06-16):**** the replay does ****NOT**** import ``windowing.py`` ? there is no ``windowing.build_windows(preset)`` / ``WindowingPreset`` / "default SERVED" call. The windowing params travel as a ****raw 3-tuple `(vt, mg, mwd)`**** pinned in each fixture's ``provenance.json`` and passed straight to ``build_candidates``; the harness deliberately avoids the named ``windowing.SERVED`` constant so a future SERVED retune can't silently churn fixtures. (``segmentation_metrics``, ``experiment.compare/Variant.spec_hash``, ``regression.save_baseline``, ``eval_stats`` are ****not**** on the M1 replay path either ? the quality-floor gate M4 will add the baseline/stats pieces.)

83

84 **### 4b. Tiered model (SEALED-FIXTURES)**

85 | Tier | Where | Key? | Contents | Test behaviour |

86 |---|---|---|---|---|

87 | ****Tier-0**** | ``eval_baselines/fixtures/`` (tracked) | none | opaque surrogate slug, fps/frame_width (retained, NOT quantized), the ****de-attributed per-frame `trajectory.csv`**** ``(Frame,Visibility,X,Y)`` (shuttle-only, time-origin reset), the 5 shuttle feature columns inside ``expected.json``, strict-slug ``label``, relative ``gts``. ****No person/audio/pose.**** ? NOT minimized to 5 scalars ? see §4c. | ****Unconditional**** ? runs on every PR/fork; turns today's skipped litmus into a real public gate |

88 | ****Tier-1**** | OneDrive (default) ****or**** SOPS-encrypted in-repo | token/age key | full corpus ****with**** person/audio; authoritative ?F1/CI/sign-test + exact Gate-A litmus | ****Skips cleanly**** when key/data absent (generalize ``_golden_present()`` ? ``_tier1_present()``) |

89

90 A fresh no-key clone is always ****green****; coverage never depends on a secret.

91

92 **### 4c. Anti-attribution measures (Tier-0)**

93

94 > ? ****RISK-ACCEPTANCE RE-CONFIRMATION REQUIRED (audit 2026-06-16).**** An earlier draft of this doc claimed Tier-0 commits "only ~5 aggregate shuttle scalars" and **"**never**** the per-frame CSV". ****That is not what ships.**** The shipped freeze (``freeze_real_fixture``) ****commits the FULL per-frame `(Frame,Visibility,X,Y)` `trajectory.csv`**** ? de-attributed (time-origin reset + opaque random slug + metadata strip), but ****not minimized to 5 scalars****. The per-frame stream is the de Montjoye fingerprint (4 points ? 95% re-ID against a published reference). The owner's 2026-06-16 risk acceptance (§7 P2) was recorded against the inaccurate "5 scalars" basis and ****MUST be RE-CONFIRMED on this corrected basis**** before any ****real**** Tier-0 fixture is committed

(P3). Synthetic fixtures (current state) carry zero provenance risk regardless.

```
95
96 - **Opaque RANDOM surrogate id** (`os.urandom` + a private surrogate?source map kept
off-git) ? *not* the MD5 `video_id`, and *not* HMAC(salt, video_id?name) (red-team: brute-
forceable over a tiny known roster if the salt leaks). Never persist the source `name`/filename.
97 - **Strict-slug `label` + dropped `source_note`** ? the operator `label` is validated
against `^[a-z0-9][a-z0-9_\-]*$` (a free-text identifier like a player/event name is refused),
and operator free-text `source_note` is **accepted-but-NOT-persisted** (never reaches the
committed `provenance.json`) ? both close the free-text identifier leak `_scan_forbidden` could
not police (audit fix #1).
98 - **Metadata strip** ? no filename, video_id, match/event/player identity, capture date.
99 - **Time-origin reset** ? subtract per-fixture `t0` (6dp-rounded) from all
start/end/gts. **Metric-invariant within `_FLOAT_TOL`** (`metrics.interval_iou` and start/end-
error are pure differences): the genuine (raw) reset-vs-no-reset score delta sits at the float-
error floor (~3.3e-7); bit-identical for whole-second offsets. See §4c-note below and §6.
100 - **Per-frame trajectory IS committed (de-attributed), NOT minimized to scalars** ?
Tier-0 commits the de-attributed per-frame `trajectory.csv` (shuttle stream only). It is **not**
reduced to ~5 aggregate scalars; the 5 shuttle *feature* columns appear (computed) inside
`expected.json`, but the raw per-frame `(Frame,Visibility,X,Y)` stream is part of the committed
fixture. (This is the corrected basis the ? note above gates on.)
101 - **Stream minimization (streams, not resolution)** ? Tier-0 keeps the shuttle block
only; person/audio excluded; pose/gait/face have **no field in the schema by design** (positive
`_scan_forbidden` assertion on the written bytes).
102 - **fps/frame_width are RETAINED as scale constants** ? they are NOT quantized today
(quantization is a deferred P3 option, see §7 ?). Reconcile any "quantize fps/frame_width"
wording elsewhere to this.
103 - **Provenance hard-fail gate** ? the de-attribution pass refuses to emit a public
fixture for any record not tagged owner-recorded(Fork-A) + minors-excluded + non-blank-license,
and refuses a negative reset label (timeline predates the trajectory). Writes go to a temp dir
and are promoted only after the privacy scan passes, so a refused freeze leaves no partial
fixture (audit fixes #3a/#3b).
104
105 ### 4d. The two gates
106 - **Characterization/snapshot** ? replay fixture, **parse-compare** JSON (never byte-
compare ? CRLF/float-format safe; `core.autocrlf=true` here), exact on int/structural fields,
`approx(abs=1e-6)` on floats; stamp + check windowing-preset/label/gt fingerprints
(incommensurable ? loud fail). Catches any behaviour change even at constant F1. **CONTROL path
only** across machines.
107 - **Quality-floor** ? score vs committed labels; assert per-video + mean F1 ? floor in
`eval_baselines/regression_baseline.json` (created here; carries a **windowing fingerprint** so
a served-default change is flagged incommensurable, not silently compared); emit `eval_stats`
paired-delta-CI + sign test (NUMBERS INFORM); add a **metrics-vs-base-branch** paired delta.
Synthetic-tier floors measure **correctness, not field quality** (documented).
108
109 ### 4e. Serialization & optional protobuf
110 JSON with `sort_keys=True` + pre-quantized floats + LF is sufficient. **Protobuf is
optional (01)** ? adopt only if cross-OS byte-stable snapshots are wanted (proto3
`rally.fixture.v1`, codegen + `protoc && git diff --exit-code` guard, `*.pb binary -text`).
Protobuf is **encoding, not secrecy** (trivially `--decode_raw`).
111
112 ### 4f. Encryption & keys (Tier-1 only)
113 **Default = out-of-repo + token** (existing OneDrive `RALLY_GOLDEN_DIR` seam ? zero
sensitive bytes in git history, sidesteps the legal "redistribution" act). **Alternative = SOPS
+ age** (per-recipient, revocable, value-level diffs) for Fork-A-clean-but-private data wanting
in-repo versioning. Rejected: git-crypt (whole-repo re-key), git-lfs (storage ? secrecy). CI
secret exposed **only to the Tier-1 job on trusted branches, never to fork-PR runners**. The
**HMAC/surrogate salt** (if used) and the surrogate?source map live with the private corpus,
rotated per release. Document explicitly that Tier-1 crypto defends **only** the no-key public.
114
115 ---
116
117 ## 5. Threat model (who's protected vs not)
118
119 | Actor | Protected against | NOT protected (by design / residual) |
120 |---|---|---|
```

```

121 | **TM1 no-key public** | map fixture?source/match/player; decrypt Tier-1; confirm a
source even holding the candidate file | read Tier-0 numbers + run shuttle-only harness
(intended; useless-in-isolation) |
122 | **TM2 keyed contributor** | recover filename/video_id/match/face/pose/gait ? *never
written* (survives the analog hole) | read every value the suite reads (unpreventable; not
pretended otherwise) |
123 | **TM3 CI runner** | leaked log can't leak attribution it never had | runs the suite
(intended); Tier-1 secret never on fork runners |
124 | **TM4 malicious keyed insider** | correlation needs a public reference; Fork-A source
is unpublished ? "not reasonably likely" | **residual:** conditional + revocable ? publish/leak
the source later and the link can snap back (forward-looking re-test duty) |
125 | **TM5 re-ID adversary** | source unpublished + reset timeline ? no back-match against
a public reference | ? becomes live if a Fork-A source is ever published ? **and the committed
per-frame trajectory IS a fingerprint** (de Montjoye 4-point re-ID), NOT just ~5 scalars
(corrected §4c). The protection is the unpublished source, not minimization |
126 | **TM6 accidental plaintext commit** | required CI leak-scan rejects MD5/known-
name/non-surrogate/plaintext-SOPS | if the required check is disabled, a leak lands |
127
128 ---
129
130 ## 6. Honest limits ? what a green suite does NOT prove
131
132 - Green proves only that the **deterministic post-trajectory transforms** are unchanged
for the **frozen cases**. It does **not** cover: the WASB/TrackNet detector,
`_probe_fps_width/cv2 frame-peek, the AI/provider handoff, chunking/re-encode, the ffmpeg
stitch, DB persistence/migrations, or the API filter. *A refactor that breaks decoding or the
detector passes these gates green.* The owner-box `regression.py` DB run remains the only full-
chain check.
133 - Characterization = **behaviour-locking, not correctness** ? it freezes current
behaviour *including current bugs*. "Passing" = "behaviour unchanged", never "behaviour
correct". This caveat must ride in the **test pass/fail message**, not only the docs.
134 - **Synthetic-tier floors = plumbing correctness, NOT field quality.** Never cite
synthetic F1 in `QUALITY_ITERATIONS.md` / the paper as rally-detection performance (tag `kind:
synthetic`).
135 - Tier-0 public coverage is **shuttle-only** ; the person/audio fusion ?F1 and exact
Gate-A values need Tier-1 (skipped on no-key clones).
136 - Data is **not "anonymous"*** absolutely (EDPS v. SRB) ? relative, conditional,
revocable, forward-looking.
137 - The committed Tier-0 stream is the **de-attributed per-frame `trajectory.csv`**, NOT
~5 scalars ? a per-frame trajectory is a de Montjoye fingerprint; the only protection is the
**unpublished source** (§4c ?, §5 TM5).
138 - **Time-origin reset is metric-invariant within `_FLOAT_TOL`, not byte-exact for every
offset.** The genuine (raw) reset-vs-no-reset score delta is at the float-error floor (~3.3e-7);
it is **bit-identical for whole-second offsets**, but because 6dp-rounding is not additive
(`round6(a)-round6(b) ? round6(a-b)`) the *serialized* 6dp boundary error can differ by one 6dp
ULP (1e-6) on a sub-second offset ? the serialization grid, not a metric divergence. (Audit fix
#2 rounds `t0`+labels to 6dp before the shift to keep the residual ? the grid step.)
139 - Encryption gives **no secrecy from a running contributor** ; protobuf gives **no
secrecy at all**.
140 - De-attribution does **not** cure provenance/licensing ? it sits *behind* the per-
record provenance gate + counsel sign-off.
141 - `--refresh-snapshot` can rubber-stamp a real regression; mitigation is **review of the
visible diff** + a separate, deliberate floor-baseline update. **A recorded `--reason` is
REQUIRED** (the CLI errors without one ? audit fix #6).
142
143 ---
144
145 ## 7. Deliverables & progress tracker ? **source of truth**
146
147 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One **small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
148
149 | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
150 |---|-----|-----|-----|-----|---|
151 | **P0** | **Leak remediation + required CI leak-scan.** Scrub real names/paths in

```

```

`eval_baselines/heuristic_lovo_n6.json`, `gemini_wrapper_2026-06-10.json`,
`backend/eval/eval_partial_labels.py`, `tests/test_cleanup_caches.py` ? opaque surrogate keys +
private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a
**required status check** ; decide on git history (PR #77, accept vs `filter-repo`). | ? | No
(owned data hygiene) | ? | ? |
152 | **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also
delivered M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema
dispatch, `Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest +
README + `.gitattributes` LF pin + `tests/test_golden_fixtures.py`. CONTROL-only (no numpy
execution on the replay path); cross-OS verified. | ? | No (synthetic) | ? |
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |
153 | **M2** | **Freeze/refresh extractor.** Core (`freeze_synthetic` / `refresh_snapshot` /
`--check` CLI + idempotence test) shipped in M1 (#189); the `--license`/`--minors-
free`/`--owned` **refusal gate** shipped with P1 (this PR). | M1 | No | ? |
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) . _this PR_ |
154 | **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared
(exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion,
perturbation + idempotence. **Shipped in M1 (#189).** (Windowing/label fingerprint guard + row-
count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
155 | **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job
(ubuntu/windows/macOS) running `--check` + the fixtures test ? locks the cross-OS guarantee
(CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No
| ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |
156 | **M4** | **Quality-floor gate + first `regression_baseline.json`.**
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
**windowing fingerprint** ; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
157 | **P1** | **De-attribution + minimization** ? `golden_fixtures.freeze_real_fixture()`
(extends the M1 module, not a separate script): **opaque random `fx_hex` surrogate slug**,
metadata strip, metric-invariant **time-origin reset**, strict GT loader, **positive no-
person/audio/MD5/source-path assertion**, manifest non-clobber; `--freeze-real` CLI with the
**refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic
tests; **no real data committed**. | M1 | No (tooling) | ? | _this PR_ |
158 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the
researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source .
minors excluded . biometric/person streams excluded . de-attributed). Owner's informed risk
acceptance ? **NOT a retained-counsel opinion** ; optional future counsel confirmation per §8. |
P1 | Owner-accepted | ? | ? |
159 | **P3** | **Tier-0 REAL subset.** Commit de-attributed, minors/biometric-excluded,
owned Fork-A real fixtures (unblocked by P2). Hard-gated **in code** by P1's refusal gate (owned
+ minors-free + license required) ? needs the owner's per-video clearance list. | P1, P2 |
conditions §3 | ? | ? |
160 | **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
161 | **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()` ; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
162 | **DOC** | **Discoverability cross-links** ? added the `eval_baselines/fixtures/` row
to `CODE_MAP §0` + a `golden_fixtures.py` entry to `CODE_MAP §2.3` ; cross-linked
`GOLDEN_DATA_SHARING.md` (committed subset ? private OneDrive corpus). README indexed in #186;
memory `current-state` kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that
doc surface is hot in the concurrent quality track; avoid collision.) | M1 | No | ? | _this PR_
|
163
164 | **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?) ; ? scrub names in
**all** files + history decision (P0) ; ? leak-scan a **required** check, hooks are convenience
only (P0) ; ? person-optional loader + **positive** no-person assertion, don't trust the silent
`0.0` default (P1 ?) ; ? random surrogate id not HMAC-over-roster (P1 ?) ; ? quantize/drop
fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option) ; ? document
Tier-0 omits the fusion path (DOC) ; ? counsel + Fork-A + minors/biometric exclusion as hard
preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).
165

```



```

166     ### Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS
job green)
167     | PR | Deliverable(s) | Suite | Coverage | Notes |
168     |----|-----|-----|-----|-----|
169     | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) | M1 +
M2-core + M3 | 746 pass / 7 skip | 80.41% | synthetic Tier-0 + CONTROL replay +
characterization; cross-OS proven (CI Linux replayed Windows-frozen fixtures) |
170     | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) | MX +
tracker/legal reconcile | full suite green | 80.41% | `golden-crossos` CI job green on
ubuntu+windows+macos ? cross-OS **enforced**; legal ? owner-accepted |
171     | _this PR_ | P1 + M2 refusal gate | 759 pass / 7 skip | **80.49%** (+0.08) |
`freeze_real_fixture` + refusal gate; **+13 tests**; tooling-only, no real data committed |
172     | _this PR (audit-hardening)_ | adversarial-audit fixes #1?#17 | 784 pass / 7 skip |
**80.71%** (+0.22) | strict-slug `label` + dropped `source_note` (#1); 6dp-rounded reset within
`_FLOAT_TOL` (#2); temp-dir-then-promote + negative-label refusal (#3); exact-membership source-
path scan (#4); key-set-equality in `compare_expected` (#5); `--reason` mandatory (#6);
independent schema guard (#7); `max_win` recorded-not-applied note (#8); numpy-execution wording
(#9); +failure-semantic & by-kind tests (#10/#11); `test_golden_real_fixtures.py` added to
`golden-crossos` CI (#12); doc reconciled to shipped reality incl. per-frame-CSV honesty + risk
re-confirm (#13?#17). Synthetic `--check` byte-identical (unchanged). |
173
174     ---
175
176     ## 8. Open owner / counsel questions
177
178     **Counsel questions ? documented basis (the owner accepted the researched risk on
2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is
optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:**
179     1. Does the EU/UK **sui-generis database right** subsist in our self-derived
shuttle/rally streams on the *Sportradar* "obtained vs created" reasoning, and does
redistributing a de-attributed Tier-0 subset cause *CV-Online* harm?
180     2. Post *Thomson Reuters v. Ross* + USCO-2025, is **US fair use for the extraction
step** defensible if the dataset could compete in a "data/analytics" market ? does publishing
only numbers (never footage) matter?
181     3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-
party ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10**
and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
182     4. Sign-off on the **final committed Tier-0 subset + provenance log**; is non-
attributability-via-private-source a defensible posture to assert given its
conditional/revocable nature?
183
184     **Owner (strategic):**
185     5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-
confidence framing) ? weighed against community/benchmark/paper aims. Your call.
186     6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source
whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the
public-video environment grows.
187     7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended)
vs a counsel-cleared real subset.
188     8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is
protobuf (01) wanted, or is canonical-JSON enough?
189
190     ---
191
192     ## 9. Definition of done
193
194     The project is **DONE** (flip `Status ? DONE`, archive) when:
195     - [ ] **P0, M1?M4, P1, DOC** are ? and merged.
196     - [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and
the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
197     - [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name /
non-surrogate / plaintext-SOPS in the tree).
198     - [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only,
windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
199     - [ ] `docs/CODE_MAP.md` §0 + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` +

```

```

memory `current-state` point here.
200     - [ ] **Either** the public commit is synthetic-only, **or** (**P2 owner-accepted ? +
P3**) for a real-data commit, under the §3 conditions enforced **in code** by P1's refusal gate.
201     - [ ] §6 honest-limits caveat rides in the test pass/fail messages, not only the docs.
202
203     *Optional 01/02 may remain ? without blocking DONE (they're earned, not assumed).*
204
205     ---
206
207     ## 10. References
208
209     **Internal code anchors (verified):** `backend/eval/{regression.py,fusion_golden.py,fusi
on_compare.py,ablation.py,experiment.py,metrics.py>windowing.py,distill_local.py,gt_loader.py,ev
al_stats.py}` · `backend/pipeline/detectors/{base.py,tracknet_runner.py,rally_gate.py}` ·
`backend/utils/hashing.py::compute_video_id` · `eval_baselines/{heuristic_lovo_n6.json,gemini_wr
apper_2026-06-10.json,experiment_leaderboard.json}` ·
`tests/test_ablation.py::test_litmus_reproduces_gate_a` · `.github/workflows/ci.yml`.
210     **Internal docs:** `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`,
`EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md
§0`.
211     **Legal authorities:** see §3 (full URLs in the research workflow artifact). *This
section is researched general information, not legal advice.*
212
213     ---
214
215     ### Changelog
216     - **2026-06-16** ? Initial draft synthesized from two multi-agent workflows (mechanics
design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified
the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.
217

```

11. assistant (2026-06-22T09:19:08.713Z)

Now let me look at the golden fixture files to understand the exact hash shapes I must NOT trip on (16-hex `fx\_` slugs, 12-hex `gt\_hash`).

12. user (2026-06-22T09:19:11.167Z)

```

README.md
clean_single_rally
manifest.json
multi_rally_with_deadtime
occlusion_break
---MANIFEST---
eval_baselines/experiment_leaderboard.json
eval_baselines/fixtures/clean_single_rally/expected.json
eval_baselines/fixtures/clean_single_rally/provenance.json
eval_baselines/fixtures/manifest.json
eval_baselines/fixtures/multi_rally_with_deadtime/expected.json
eval_baselines/fixtures/multi_rally_with_deadtime/provenance.json
eval_baselines/fixtures/occlusion_break/expected.json
eval_baselines/fixtures/occlusion_break/provenance.json
eval_baselines/gemini_wrapper_2026-06-10.json
eval_baselines/heuristic_lovo_n6.json
eval_baselines/nightly_baseline.json

```

13. user (2026-06-22T09:19:11.858Z)

```

eval_baselines\fixtures\clean_single_rally\expected.json:21:  "gt_hash": "6c93801fd607",
eval_baselines\fixtures\occlusion_break\expected.json:36:  "gt_hash": "bbec003aed74",
eval_baselines\fixtures\multi_rally_with_deadtime\expected.json:36:  "gt_hash": "bbec003aed74",

```

14. assistant (2026-06-22T09:19:15.457Z)

Let me look at the provenance.json (which holds the `fx\_` slug) and check for any 32-hex tokens or personal paths currently in the scan scope that I should be aware of.

15. user (2026-06-22T09:19:16.426Z)

```
1      {
2      "description": "One continuous ~4s rally; shuttle oscillates across the net ? 1
window.",
3      "fps": 30.0,
4      "frame_width": 1280.0,
5      "iou": 0.5,
6      "kind": "synthetic",
7      "license": "synthetic",
8      "minors_free": true,
9      "note": "Synthetic, deterministic, owned (no real footage). Tier-0 / CONTROL-only.
Plumbing-correctness fixture ? NOT a measure of rally-detection quality (§6).",
10     "paths": {
11         "expected": "expected.json",
12         "labels": "labels.csv",
13         "trajectory": "trajectory.csv"
14     },
15     "schema_version": 1,
16     "slug": "clean_single_rally",
17     "windowing": {
18         "max_win": 0.0,
19         "mg": 2.0,
20         "mwd": 2.0,
21         "vt": 0.02
22     }
23 }
24
```

16. user (2026-06-22T09:19:17.468Z)

(Bash completed with no output)

17. assistant (2026-06-22T09:19:20.767Z)

The synthetic fixtures use human-readable slugs (no `fx\_` hex). The `fx\_` slugs are for real fixtures (none committed yet). The `gt\_hash` is 12-hex. Let me now run a quick scan to discover what 32-hex tokens or personal paths currently exist in the scan scope, so I know what the scanner will flag.

18. user (2026-06-22T09:19:22.378Z)

```
=== git-tracked files in scan roots ===
315
```

19. user (2026-06-22T09:19:36.210Z)

```
=== 32-hex tokens in scan roots (tracked) ===
tests/test_golden_real_fixtures.py:467:0123456789abcdef0123456789abcdef
tests/test_workspace.py:21:27f1c707bbfb0a1b2c3d4e5f60718293
```

20. assistant (2026-06-22T09:19:38.600Z)

There are two existing 32-hex tokens in test files. Let me look at these to understand their context, then check for personal paths in the scan roots.

21. user (2026-06-22T09:19:40.026Z)

```
455         assert mism and any(m.startswith("score:") for m in mism), mism
456
457
```

```

458     def test_scan_forbidden_raises_on_forbidden_key():
459         """red-team #10: the FORBIDDEN_KEYS branch raises (a person/audio/pose stream in the
text)."""
460         with pytest.raises(gf.RefusalError) as ei:
461             gf._scan_forbidden({'person': [1,2,3]}', where="x", source_paths=[])
462             assert "forbidden key" in str(ei.value)
463
464
465     def test_scan_forbidden_raises_on_md5():
466         """red-team #10: the MD5 branch raises (a 32-hex video_id shape is an attribution
back-channel)."""
467         md5 = "0123456789abcdef0123456789abcdef"
468         with pytest.raises(gf.RefusalError) as ei:
469             gf._scan_forbidden({'slug': "%s" % md5, where="x", source_paths=[])
470             assert "MD5" in str(ei.value)
471
472
473     def test_refresh_snapshot_no_target_raises(tmp_path):
474         """red-team #10: refresh_snapshot with neither slug nor all raises ValueError."""
475         with pytest.raises(ValueError):
476             gf.refresh_snapshot(fixtures_dir=tmp_path)
477
478
479     def test_refresh_snapshot_cli_requires_reason(capsys):

```

22. user (2026-06-22T09:19:40.693Z)

```

10
11     from backend.utils.workspace import (
12         LOCAL_ONLY_SUBDIRS,
13         SUBDIRS,
14         VideoWorkspace,
15         for_video,
16         join_uri,
17         local_scratch_base,
18         resolve_workspace_root,
19     )
20
21     VID = "27f1c707bbfb0a1b2c3d4e5f60718293" # MD5-shaped video_id
22
23
24     class TestJoinUri:
25         def test_basic_join(self):
26             assert join_uri("gs://bucket", "workspaces", VID) ==
f"gs://bucket/workspaces/{VID}"
27
28         def test_strips_trailing_and_inner_slashes(self):
29             assert join_uri("gs://bucket/", "/a/", "/b/") == "gs://bucket/a/b"
30
31         def test_skips_blank_parts(self):
32             assert join_uri("root", "", "x", " ", "y") == "root/x/y"
33             assert join_uri("root") == "root"
34             assert join_uri("root", "") == "root"

```

23. assistant (2026-06-22T09:19:50.576Z)

Both are legitimate synthetic MD5-shaped test fixtures (`0123456789abcdef...` and `27f1c707...`), not real video_ids. They are test data deliberately demonstrating the MD5 shape. The scanner as specified would flag these ? but the spec says to exclude only `tools/leak\_scan.py` itself and its test file.

This is a real conflict: the MD5 regex `b[0-9a-fA-F]{32}b` will trip on these two test files. The spec says "Exit 1 if any finding". Let me reconsider ? these are synthetic test fixtures, not real leaks. The scanner must exit 0 (clean) on the current tree per VERIFY step 1.

I need an allowlist mechanism for these legitimate synthetic test MD5s. The cleanest, most honest approach: support an inline allow-comment marker (e.g. `# leak-scan: allow` or `noqa`-style) that lets a line opt out, and add it to these two test lines. This keeps the guard strict by default but lets known-synthetic test fixtures be marked. Let me check whether these test files have other constraints, and check the broader picture: also confirm there are no personal paths lurking elsewhere in scope.

24. user (2026-06-22T09:20:05.217Z)

```
=== personal C:/Users/<name> or /home|Users/<name>/ in tracked scan-root files ===
backend/eval/eval_partial_labels.py:23:C:/Users/avidu
backend/eval/eval_partial_labels.py:24:C:/Users/avidu
backend/pipeline/detectors/tracknet_runner.py:64:C:\\Users\\x
scripts/register_nightly_task.ps1:17:C:\\Users\\avidu
tests/test_p2b_install.py:91:/home/u/
tests/test_p2b_install.py:101:/home/u/
tests/test_p2b_install.py:128:C:\\Users\\u
tests/test_p2b_install.py:130:/home/u/
tests/test_p2b_install.py:131:/home/u/
tests/test_p2b_install.py:135:/home/u/
tests/test_storage.py:26:C:/Users/a
tests/test_storage.py:26:C:/Users/a
tests/test_storage.py:27:C:\\Users\\a
tests/test_storage.py:27:C:\\Users\\a
tests/test_tracknet_runner.py:26:C:\\Users\\avidu
tests/test_tracknet_runner.py:26:/Users/avidu/
tests/test_tracknet_runner.py:32:/home/avidullu/
tests/test_tracknet_runner.py:32:/home/avidullu/
tests/test_wsl_tilde_quoting.py:22:/Users/x/
tests/test_wsl_tilde_quoting.py:22:/Users/x/
```

25. assistant (2026-06-22T09:20:09.638Z)

This is important. There are several matches, but I must distinguish real personal usernames from short placeholder-like names. Let me examine each. The spec says flag a path where `` is a real username, NOT a placeholder. The allowlist given is ``, ``, ``, `you`, `youruser`, `yourname`. But there are also short synthetic names like `x`, `u`, `a` used in tests. Let me read the actual context of each.

26. user (2026-06-22T09:20:12.912Z)

```
===== backend/pipeline/detectors/tracknet_runner.py (60-68) =====
    )

def to_wsl_mnt_path(win_path: str) -> str:
    """Translate a Windows path (C:\\Users\\x) to its WSL /mnt form (/mnt/c/Users/x).

    Already-POSIX paths (starting with / or ~) are returned unchanged so the
    same helper is safe to call on either kind of path.
    """
===== scripts/register_nightly_task.ps1 (13-21) =====
    is caught up on wake.

.EXAMPLE
    # Register against the main checkout, nightly at 03:00:
    pwsh -File scripts/register_nightly_task.ps1 -RepoRoot 'C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer'

.EXAMPLE
    # Remove it:
    pwsh -File scripts/register_nightly_task.ps1 -Unregister
===== tests/test_p2b_install.py (88-135) =====
    ("Linux", "khelsutra.sh", True)],
```

```

)
def test_shortcut_spec_per_os(install_mod, system, fname, executable):
    home = Path("/home/u/AppHome")
    name, content, ex = install_mod.shortcut_spec(system, home)
    assert name == fname and ex is executable
    # str(home) is OS-native (backslashes on Windows) ? assert against it, not a hardcoded
    literal.
    assert "RALLY_APP_HOME" in content and str(home) in content
    assert "indexer" in content and "--app" in content # command is quoted ("indexer" --app)

def test_shortcut_spec_embeds_quoted_command(install_mod):
    """An absolute indexer path (resolved at install time, may contain spaces) is embedded +
    quoted."""
    _, content, _ = install_mod.shortcut_spec("Linux", Path("/home/u/AppHome"), "/opt/uv
bin/indexer")
    assert "'/opt/uv bin/indexer' --app" in content

def test_script_app_home_matches_engine(install_mod, uninstall_mod):
    """The stdlib mirrors in scripts/ must not drift from the engine's resolver (gate vs silent
    state-split): same APP_DIR_NAME, and same app-home on the current platform."""
    from backend.utils import app_home as engine_ah
    assert install_mod.APP_DIR_NAME == engine_ah.APP_DIR_NAME == uninstall_mod.APP_DIR_NAME
    assert install_mod.default_os_app_home() == engine_ah.default_os_app_home()
    assert uninstall_mod.default_os_app_home() == engine_ah.default_os_app_home()

def test_resolve_method(install_mod):
    assert install_mod.resolve_method("auto", uv_present=True) == "uv"
    assert install_mod.resolve_method("auto", uv_present=False) == "pip"
    assert install_mod.resolve_method("pip", uv_present=True) == "pip" # explicit wins
    assert install_mod.resolve_method("uv", uv_present=False) == "uv"

def test_install_command_argv(install_mod):
    assert install_mod.install_command("uv", ".") == ["uv", "tool", "install", "."]
    pip = install_mod.install_command("pip", "badminton-highlight-indexer")
    assert pip[1:] == ["-m", "pip", "install", "badminton-highlight-indexer"]

def test_default_os_app_home_per_os(install_mod):
    win = install_mod.default_os_app_home(system="Windows", env={"APPDATA":
r"C:\Users\u\AppData\Roaming"})
    assert win.name == "Khelsutra" and "Roaming" in str(win)
    lin = install_mod.default_os_app_home(system="Linux", env={"XDG_DATA_HOME":
"/home/u/.local/share"})
    assert str(lin) == str(Path("/home/u/.local/share") / "Khelsutra")

def test_build_manifest_shape(install_mod):
    home = Path("/home/u/AppHome")
    ===== tests/test_storage.py (24-30) =====
    ("/data/x.mp4", "local", "", "/data/x.mp4"),
    ("rel/x.mp4", "local", "", "rel/x.mp4"),
    ("C:/Users/a/x.mp4", "local", "", "C:/Users/a/x.mp4"), # Windows path: no '://'
    ("C:\\Users\\a\\x.mp4", "local", "", "C:\\Users\\a\\x.mp4"),
    ("local:///abs/x.mp4", "local", "", "/abs/x.mp4"),
    ("s3://rally/videos/m.mp4", "s3", "rally", "videos/m.mp4"),
    ("gs://rally-eu/w/model.json", "gcs", "rally-eu", "w/model.json"), # gs:// -> gcs
    ===== tests/test_tracknet_runner.py (24-36) =====

def test_to_wsl_mnt_path_windows():
    assert to_wsl_mnt_path(r"C:\Users\avidu\v.mp4") == "/mnt/c/Users/avidu/v.mp4"
    assert to_wsl_mnt_path(r"D:\clips\m.mp4") == "/mnt/d/clips/m.mp4"

```

```

def test_to_wsl_mnt_path_passthrough():
    # Already-POSIX paths are left alone.
    assert to_wsl_mnt_path("/home/avidullu/clips/v.mp4") == "/home/avidullu/clips/v.mp4"
    assert to_wsl_mnt_path("~/clips/v.mp4") == "~/clips/v.mp4"

def test_config_from_indexing_cfg():
===== tests/test_wsl_tilde_quoting.py (18-26) =====
@pytest.mark.parametrize("path,expected", [
    ("~/clips/x.mp4", "~/clips/x.mp4"),          # safe chars: quote is a no-op, ~ expandable
    ("~/clips/has space.mp4", "~/clips/has space.mp4"),
    ("~", "~"),                                  # bare home
    ("/mnt/c/Users/x/out", "/mnt/c/Users/x/out"), # shlex.quote no-op for safe paths
    ("/mnt/c/has space", "'/mnt/c/has space'"),   # ordinary quoting preserved
])
def test_wsl_tilde_quote(path, expected):
    assert wsl_tilde_quote(path) == expected

```

27. assistant (2026-06-22T09:20:34.330Z)

This is very informative. The findings split into two clear categories:

****Real personal-username leaks**** (the actual `avidu`/`avidullu` username ? these ARE leaks):

- `backend/pipeline/detectors/tracknet\_runner.py:64` ? `C:\Users\x` ? this is `x`, a PLACEHOLDER (short, illustrative). Not a leak.
- `scripts/register\_nightly\_task.ps1:17` ? `C:\Users\avidu\Projects\...` ? REAL username `avidu`. This is in `EXAMPLE` docstring. ****But the task says touch ONLY the listed files.**** This is NOT in my list.
- `tests/test\_tracknet\_runner.py:26,32` ? `avidu`/`avidullu` ? REAL usernames, but in test assertions for WSL path translation. NOT in my list.

****Placeholders / short synthetic names**** (`x`, `u`, `a`): these are NOT real usernames. The scanner must not flag them.

This is the core design tension. The spec's regex ``<name>`` is "a real username, i.e. NOT one of an allowlist of placeholders". The given allowlist is ``<name>``, ``<user>``, ``<username>``, ``you``, ``youruser``, ``yourname``. Single letters `x`, `u`, `a` are NOT in that allowlist but are clearly placeholders.

If I implement the regex literally, the scanner would flag `register\_nightly\_task.ps1:17` (`avidu`), `test\_tracknet\_runner.py:26,32` (`avidu`/`avidullu`) ? files I am told NOT to touch and which are NOT the deferred name-rename either. The VERIFY step says the scan MUST exit 0 after my scrub of the listed files.

Let me re-read the VERIFY instruction carefully. It says: ****"If it flags something you were told NOT to scrub (e.g. a name in eval_baselines), that means the scan scope is too broad ? NARROW the scan to not flag the DEFERRED names (it should only flag MD5 + personal paths, not the video names)."** That guidance is about ****video names**** (functional keys like "mahadevpura-1"). It does NOT say to suppress real personal paths like `avidu` in `register\_nightly\_task.ps1`.

This is a genuine conflict that I cannot resolve by silently weakening the scanner (that would defeat the guard's purpose) nor by editing out-of-scope files. Let me think about the right design.

The honest path: the scanner needs an ****inline allow-marker**** mechanism (like ``# noqa``) so that legitimate occurrences ? synthetic test MD5s, real-username test assertions that test path translation, the example in the PS1 script ? can be explicitly annotated. But adding markers to `register\_nightly\_task.ps1` / `test\_tracknet\_runner.py` would mean editing files outside my list.

Let me reconsider the scope boundary. The instruction says "Touch ONLY the files listed below" for the ****scrub**** (section 1). But the scanner must pass clean. Let me check: are these out-of-scope personal-path files perhaps already intended to be scrubbed in a different way, or is the regex meant to be narrower? Let me look more carefully ? the key distinction the spec draws is real-username vs placeholder. Single letters and the test-context names may need a different

treatment. Let me check git history / other sessions for how this was intended. Let me re-examine: the spec explicitly lists which files to scrub and says the scan must be clean. The most defensible reading: the scanner should use an ****inline-allow comment**** for legitimate cases, AND I should add that marker to the minimum set of lines needed. But that touches out-of-scope files.

Let me look at whether there's an even narrower path: perhaps the personal-path regex should require a path with enough structure (e.g. `Projects`, a real directory) OR perhaps these test files should be excludable. Let me reconsider the design with the allow-marker as the clean solution, and check exactly which lines would trip.

28. user (2026-06-22T09:20:35.916Z)

```
1      <#
2      .SYNOPSIS
3          Register (or remove) the nightly rally-quality regression as a Windows Scheduled
Task.
4
5      .DESCRIPTION
6          Creates a daily user-level task that runs scripts/nightly_regression.ps1 against a
repo
7          checkout. The action is inline-guarded, so it's safe to register BEFORE this work is
merged
8          into the target checkout: it no-ops + logs "awaiting" until the wrapper exists
there, then
9          self-activates. Idempotent (-Force re-registers); reversible (-Unregister).
10
11         Registers under the current user (no admin / no -RunLevel Highest needed ?
consistent with the
12         repo's other scheduled tasks). Uses -StartWhenAvailable so a run missed while the
machine slept
13         is caught up on wake.
14
15     .EXAMPLE
16         # Register against the main checkout, nightly at 03:00:
17         pwsh -File scripts/register_nightly_task.ps1 -RepoRoot
'C:\Users\avidu\Projects\badminton-highlight-indexer'
18
19     .EXAMPLE
20         # Remove it:
21         pwsh -File scripts/register_nightly_task.ps1 -Unregister
22     #>
23     [CmdletBinding()]
24     param(
25         [string]$RepoRoot = (Split-Path -Parent (Split-Path -Parent
$MyInvocation.MyCommand.Path)),
26         [string]$TaskName = 'KhelSutraNightlyRegression',
27         [string]$Time = '03:00',
28         [switch]$Pull,
29         [switch]$Unregister
30     )
```

29. assistant (2026-06-22T09:20:37.652Z)

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